

GAUGES

Limit gauging

Adoption of a system of limits and fits logically leads to the use of limit gauges, with which no attempt is made to determine the size of a workpiece - they are simply used to find whether the component is within the specified limits of size or not. The simplest form of limit gauges are those used for inspecting holes or shafts.

Unfortunately it is not as simple as this, for the same reason that limits of size are required for the work; nothing can be made to an exact size and this includes gauges. Thus the gauge maker requires a tolerance to which he may work, and the positioning of this gauge tolerance relative to the nominal gauge size requires a policy decision. For instance, if the gauge tolerance increases the size of a 'GO' plug gauge, and decreases the size of the 'NOT GO' end, the gauge will tend to reject good work which is near the upper or lower size limits.

Similarly if the gauge

In , where a large number of similar components are manufactured on an interchangeable basis, measuring the dimensions of each part will be a time-consuming and expensive exercise. In addition, the actual or absolute size of a component, provided that it is within the limits specified, is not of much importance because the permissible limits of variations in dimensions would have been specified by the designer in a logical manner, giving due consideration to the functional requirements. Therefore, in mass production, gauges can be used to check for the compliance of the limits of the part with the permissive tolerance limits, instead of measuring the actual dimensions. The term 'limit gauging' signifies the use of gauges for checking the limits of the components. Gauging plays an important role in the control of dimensions and interchangeable manufacture.

mass production  measuring the dimensions time-consuming
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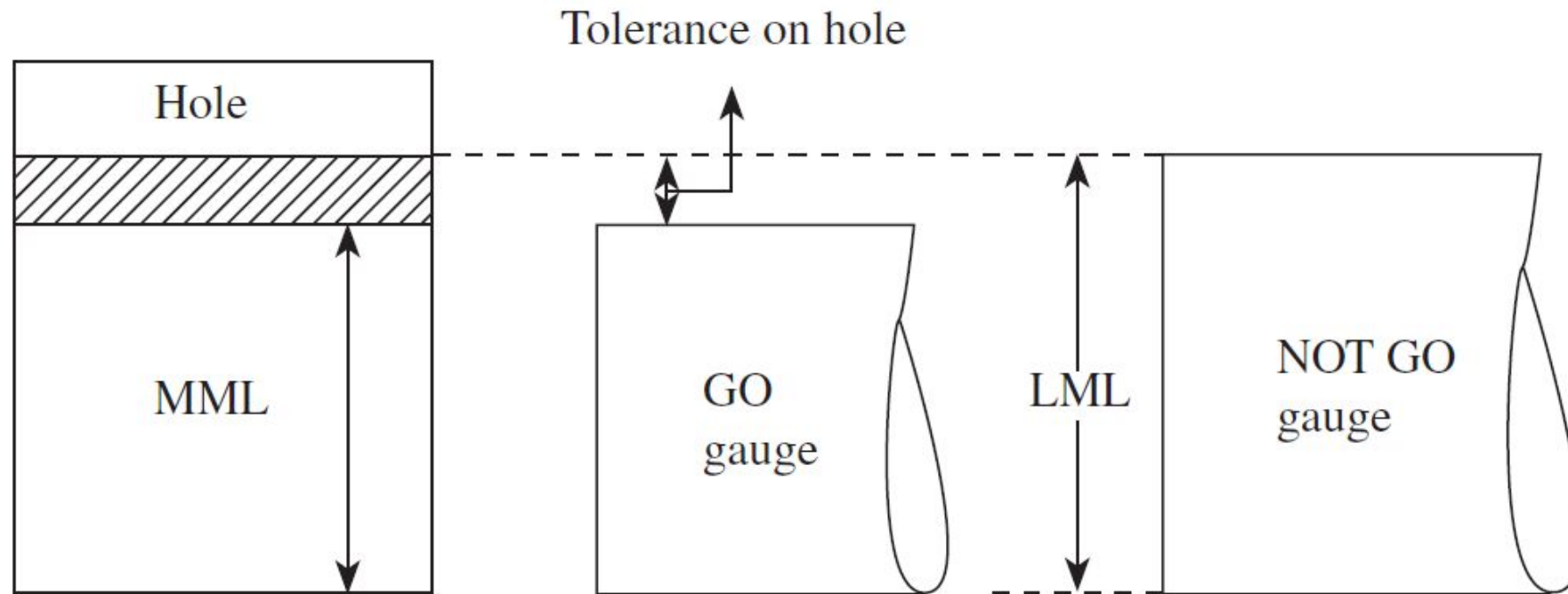
Gauges  check for the compliance of the limits of the
part with the permissive tolerance limits, instead
of measuring the actual dimensions

Consider first a hole on which the limits on diameter are specified. The 'GO' gauge is a cylinder whose diameter is equal to the minimum hole size, and that the 'NOT GO' gauge is a similar cylinder equal in diameter to the maximum hole size.

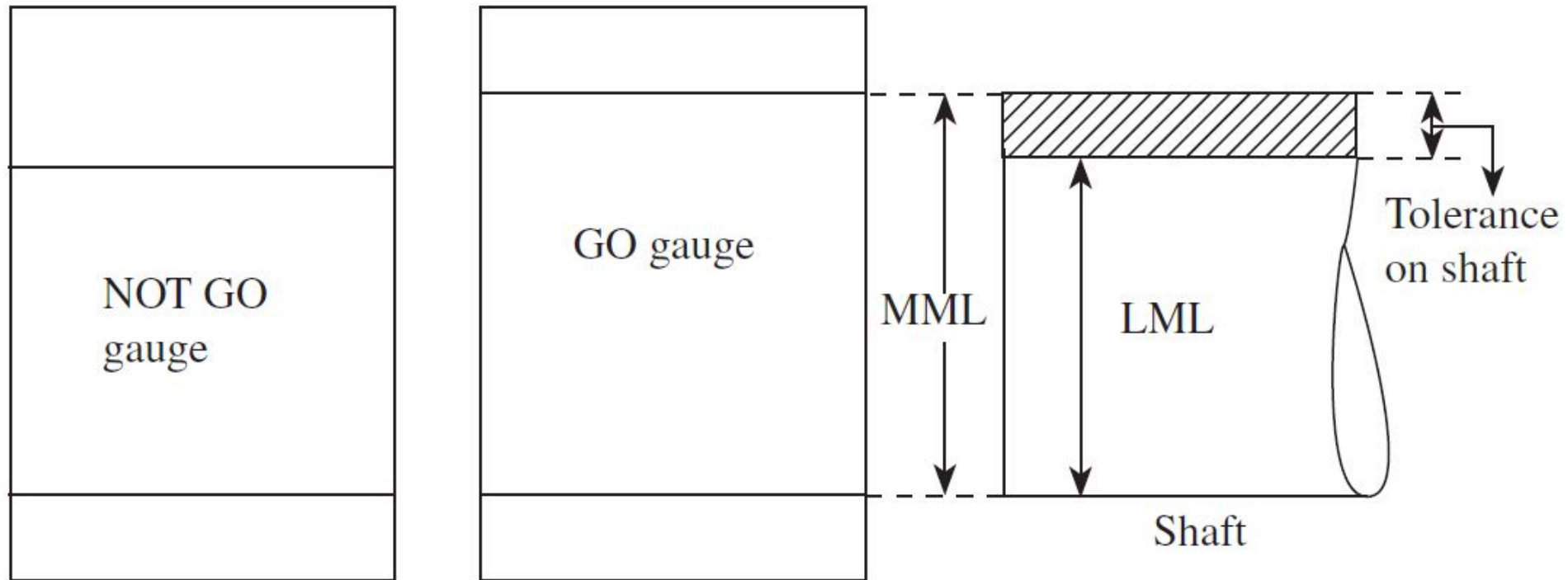
GO gauges  MML

NO GO or NOT GO gauges  LML

Metal limits for hole gauging



Metal limits for shaft gauging



Clearance fits are preferred for a majority of the assembly operations .

Allowance or minimum clearance is determined by the algebraic difference of the MMLs of the mating parts. Therefore, for clearance fits, the MMLs of the mating parts become more critical than the LMLs.

Whenever the components are gauged for their MMLs, if the GO gauges fail to assemble during inspection, the components should not be accepted under any circumstances.

Classification of Gauges

1. Plain gauges

(a) According to their type: (i) Standard gauges (ii) Limit gauges

(b) According to their purpose: (i) Workshop (ii) Inspection (iii) Reference, or master, or control gauges

(c) According to the form of the tested surface:

(i) Plug gauges for checking holes (ii) Snap and ring gauges for checking shafts

(d) According to their design:

(i) Single- and double-limit gauges

(ii) Single- and double-ended gauges

(iii) Fixed and adjustable gauges

2. Adjustable-type gap gauges

3. Miscellaneous gauges

(a) Combined-limit gauges

(b) Taper gauges

(c) Position gauges

(d) Receiver gauges

(e) Contour gauges

(f) Profile gauges

Taylor's Principle

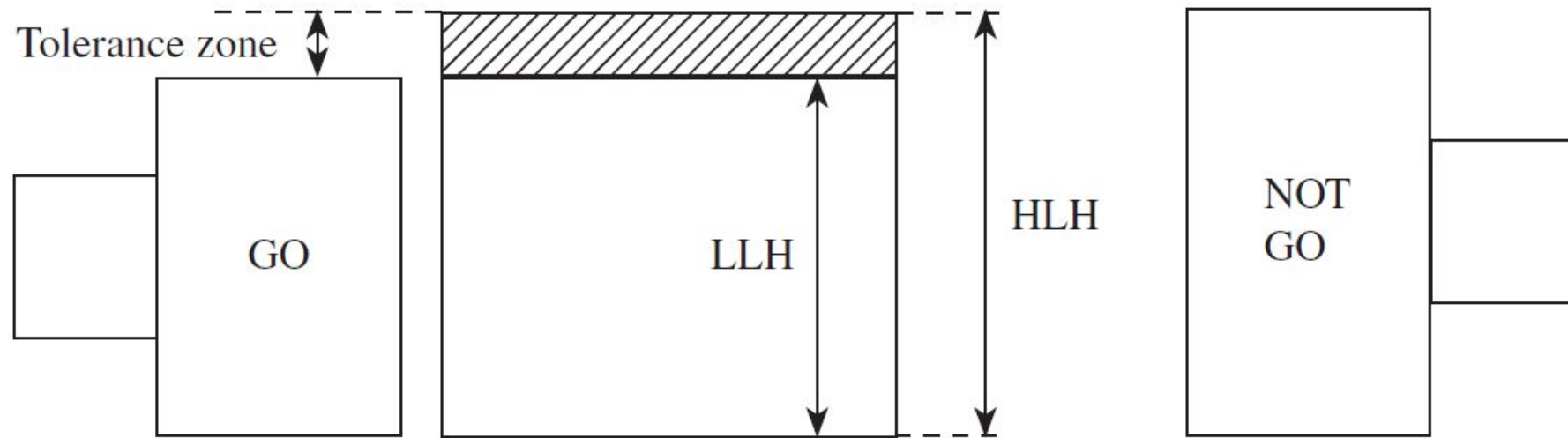
Taylor's principle states that

The GO gauge is designed to check maximum metal conditions, that is, LLH and HLS. It should also simultaneously check as many related dimensions, such as roundness, size, and location, as possible.

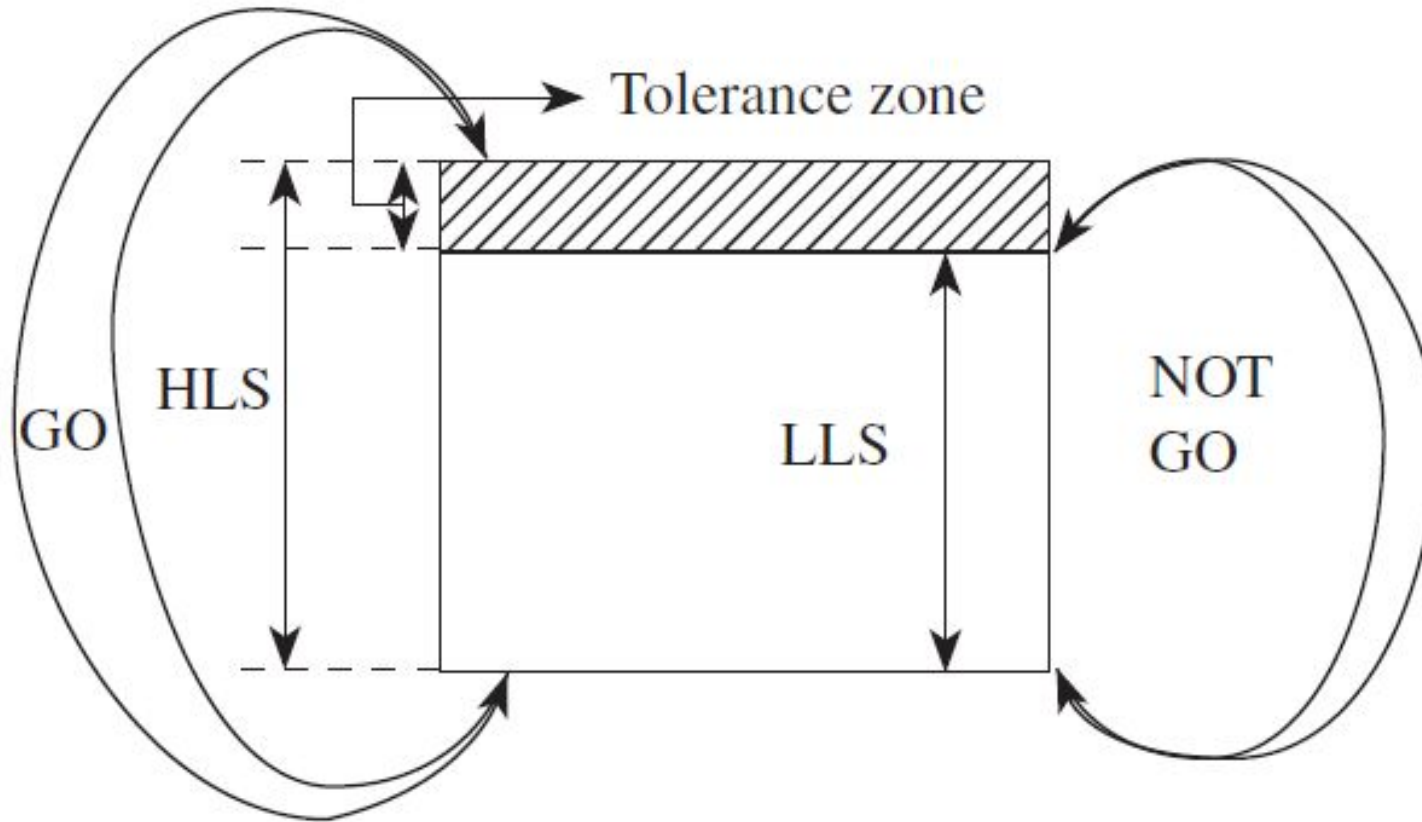
The NOT GO gauge is designed to check minimum metal conditions, that is, HLH and LLS. It should check only one dimension at a time. Thus, a separate NOT GO gauge is required for each individual dimension.

During inspection, the GO side of the gauge should enter the hole or just pass over the shaft under the weight of the gauge without using undue force. The NOT GO side should not enter or pass.

GO and NOT GO limits of plug gauge



GO and NOT GO limits of snap gauge



The GO plug is used to check more than one dimension of the hole simultaneously, the GO plug gauge must have a full circular section and must be of full length of the hole so that straightness of the hole can also be checked. During inspection, it can be ensured that if there is any lack of straightness or roundness of the hole a full entry of the GO plug gauge will not be allowed. Thus, it not only controls the diameter in any given crosssection but also ensures better bore alignment. However, it should be mentioned here that the GO plug gauge cannot check the degree of ovality.

Important Points for Gauge Design

1. The form of GO gauges should be a replica of the form of the opposed (mating) parts.
2. GO gauges enable several related dimensions to be checked simultaneously and hence are termed complex gauges.
3. During inspection, GO gauges must always be put into conditions of maximum impassability.
4. NOT GO gauges check a single element of feature at a time.
5. In inspection, NOT GO gauges must always be put into conditions of maximum passability.

Gauge Tolerance (Gauge Maker's Tolerance)

Thus, the tolerance that is allowed in the manufacture of gauges is termed gauge maker's tolerance or simply gauge tolerance. Logically, gauge tolerance should be kept as minimum as possible; however, this increases the gauge manufacturing cost. The normal practice is to take gauge tolerance as 10% of the work tolerance.

Wear Allowance

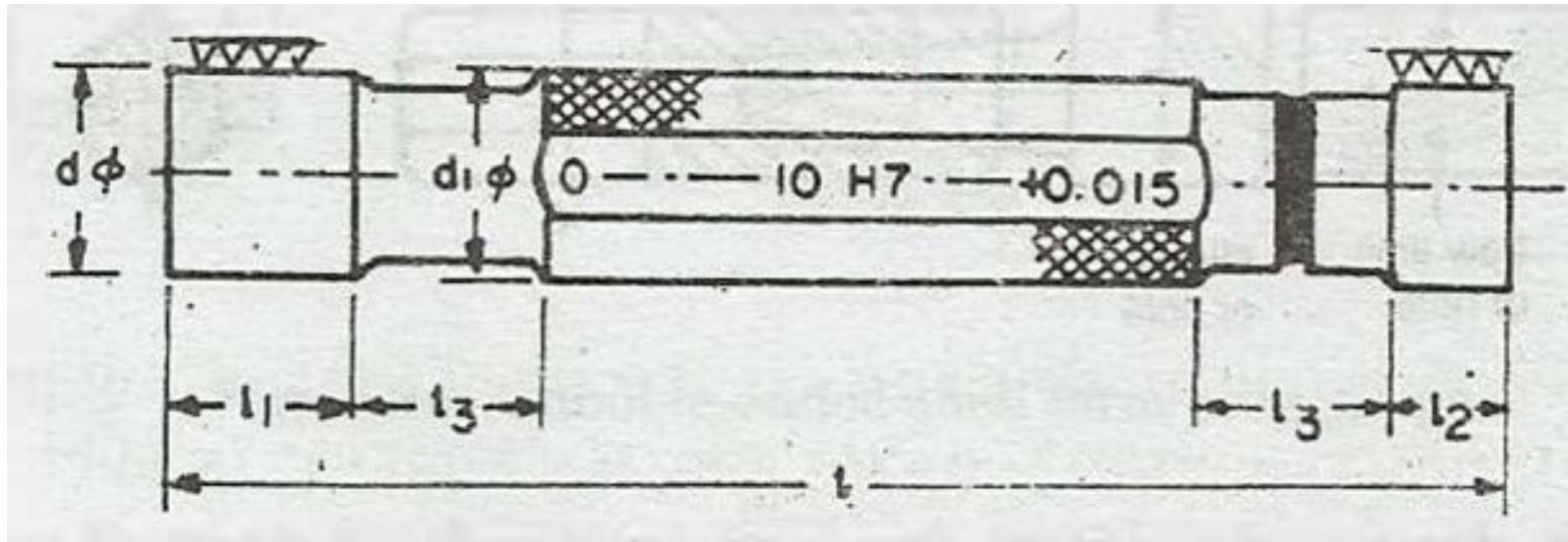
According to Taylor's principle, during inspection the NOT GO side should not enter or pass. The NOT GO gauge seldom engages fully with the work and therefore does not undergo any wear. Hence, there is no need to provide an allowance for wear in case of NOT GO gauges.

Taylor's principle also states that the GO side of the gauge should enter the hole or just pass over the shaft under the weight of the gauge without using undue force. During inspection, the measuring surfaces of the gauge constantly rub against the mating surfaces of the workpiece. Therefore, the GO gauges suffer wear on the measuring surfaces and thus lose their initial dimension. Hence, wear allowance is provided for GO gauges to extend their service life. As a consequence of this wear, the size of the GO plug gauge decreases while that of the ring or gap gauge increases. The wear allowance provided for the GO gauges are added in a direction opposite to wear. This allowance is added in for a plug gauge while subtracted for a ring or gap gauge. A wear allowance of 10% of gauge tolerance is widely accepted in industries. If the work tolerance of a component is less than 0.1 mm, no wear allowance on gauges is provided for that component, since a wear allowance of less than 0.001 mm will not have any practical effect on the gauges.

The basic or nominal size of the GO side of the gauge conforms to the LLH or HLS, since it is designed to check maximum metal conditions. In contrast, the basic or nominal size of the NOT GO gauge corresponds to HLH or LLS, as it is designed to check minimum metal conditions.

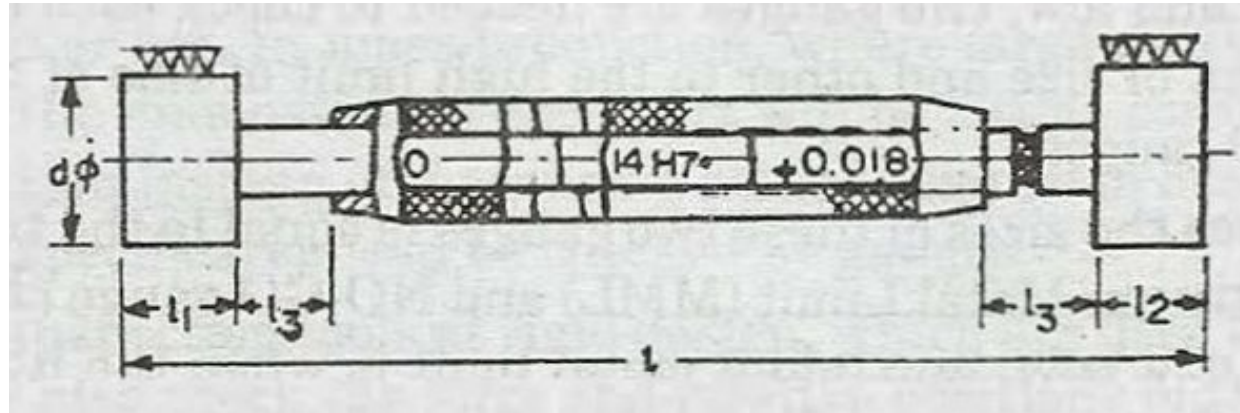
Types of Plug Gauges

1. Solid type (For sizes up to 10 mm)



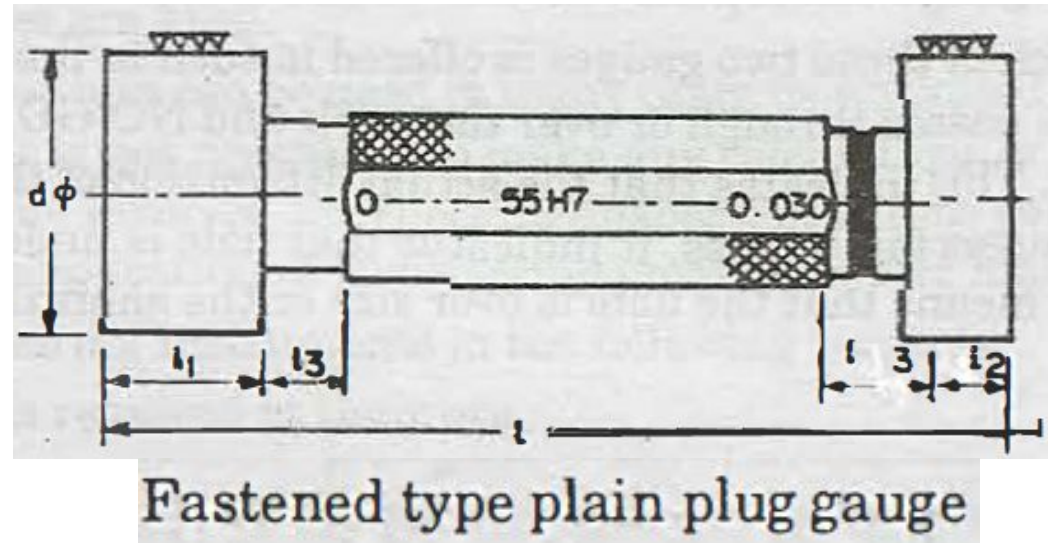
Solid type double-ended plain plug gauge

2. Renewable type (Taper inserted type). For sizes over 10 mm and up to 30 mm



Inserted type plain plug gauge

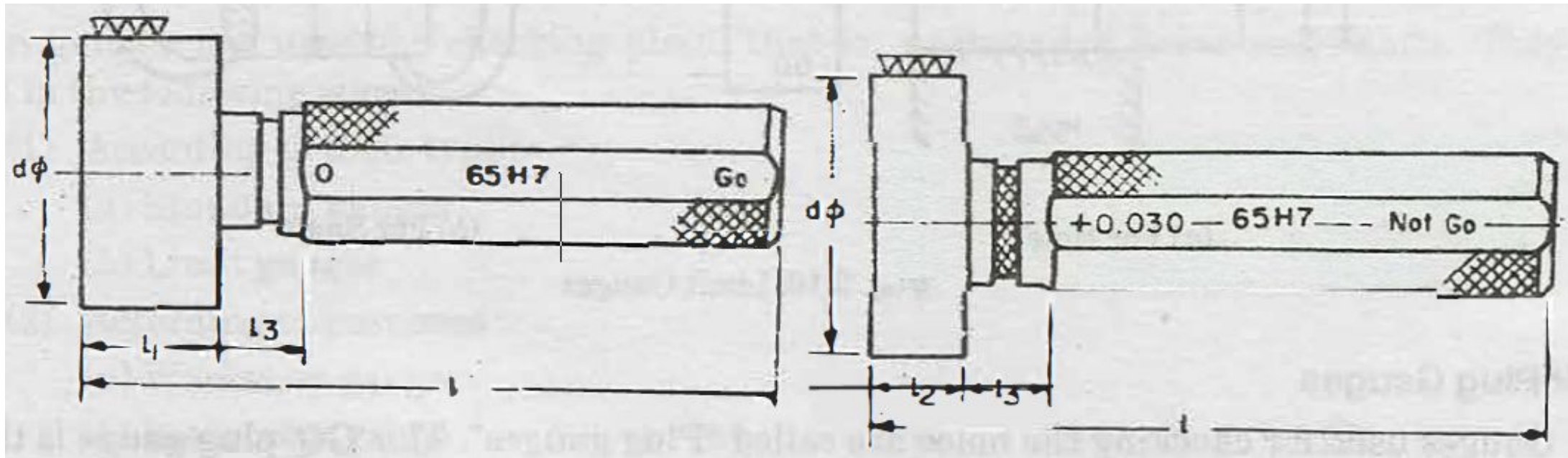
3. Fastened type



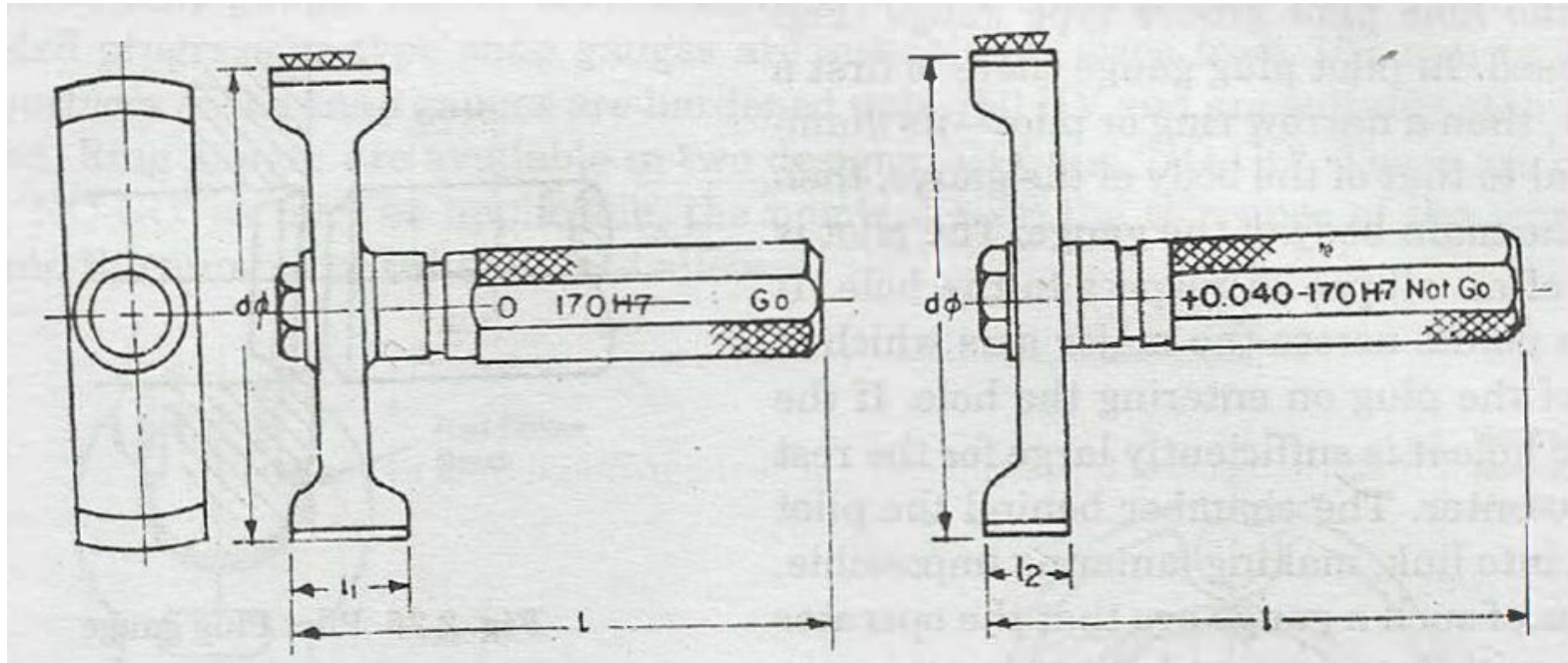
(a) **Double-ended** : For sizes over 30 mm and up to 63 mm

(b) **Single-ended**: For sizes over 63 mm and up to 100 mm

Fastened type Go and Not-Go plug single-ended gauges

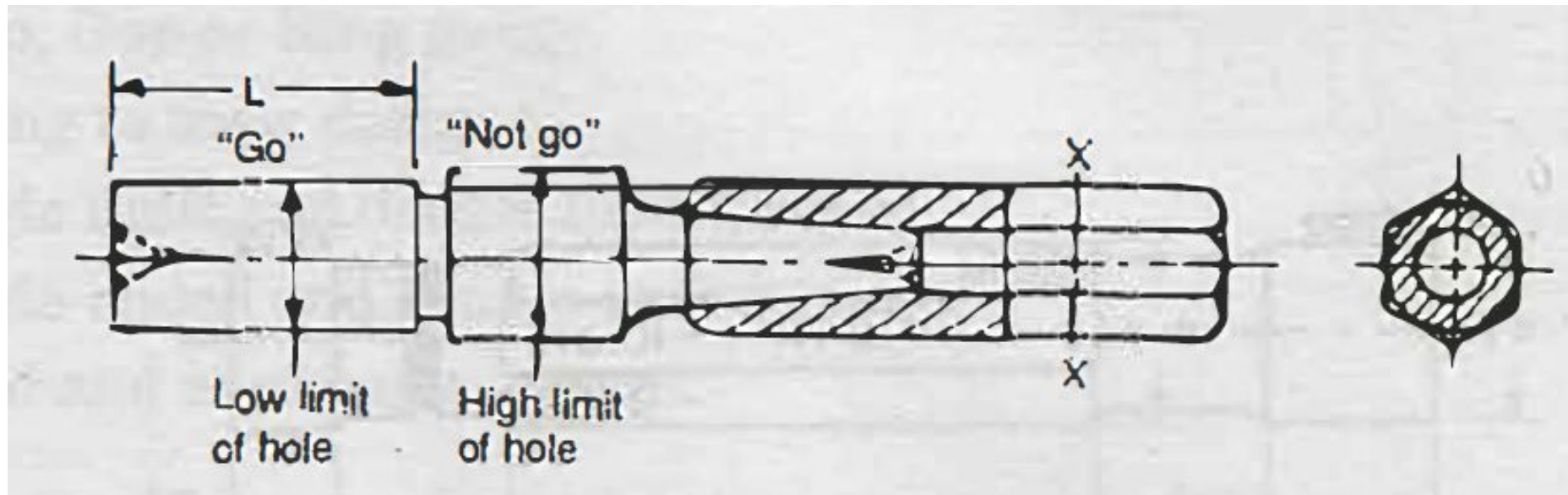


4. Flat type. For sizes over 100 mm and up to 250 mm.

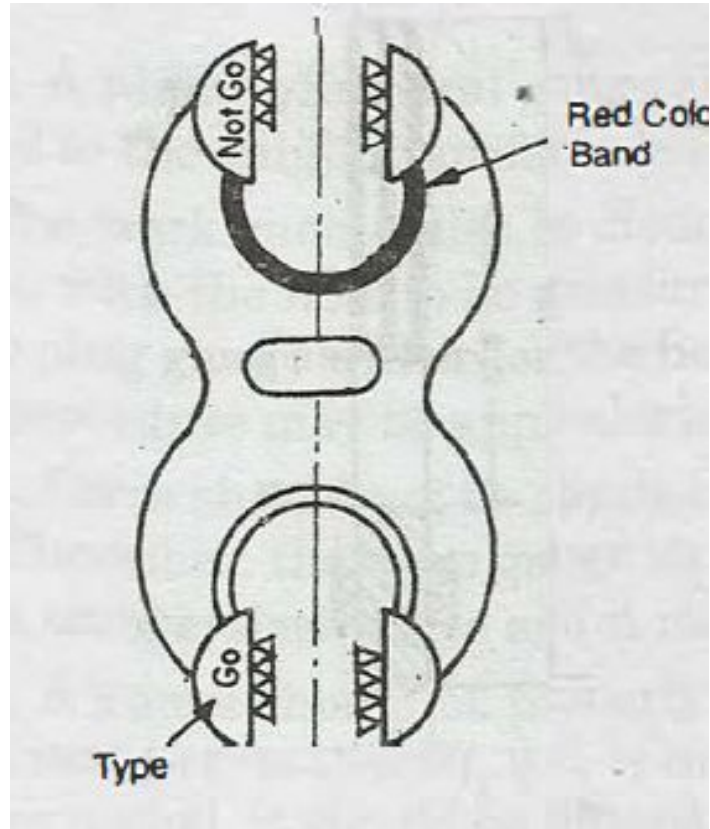


Flat type Go and Not-Go plug single-ended gauges

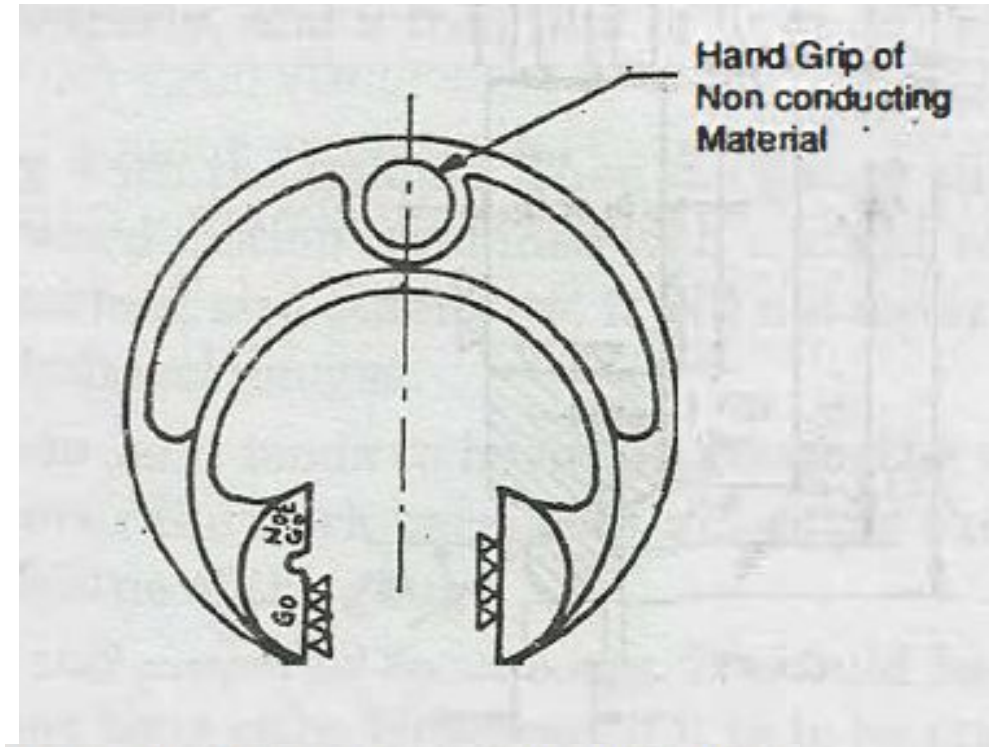
5. Progressive type. For relatively short through hole. It has both the ends on one side of the gauge



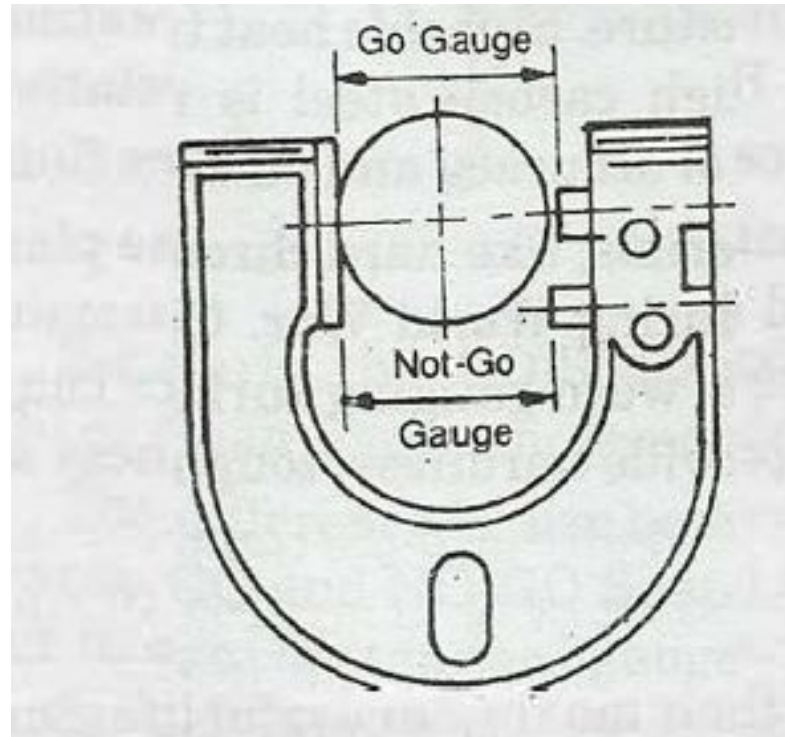
Snap, Gap or Ring Gauges



Double-ended snap gauge



Progressive type snap gauge for sizes above



Adjustable gap gauge